IP Addressing and Subnetting

1. IP Address Classes (A, B, C, D, E)

Class	First Octet Range	Default Subnet Mask	Real-World Example
Class A	1–126	255.0.0.0 (/8)	A large organization's private network using 10.0.0.1.
Class B	128–191	255.255.0.0 (/16)	A college network using 172.16.0.1.
Class C	192–223	255.255.255.0 (/24)	A home router using 192.168.1.1.
Class D	224–239	Not applicable	Multicast streaming or video distribution.
Class E	240–255	Not applicable	Reserved for experimental or future use.

Note: 127.x.x.x is reserved for loopback testing. Classes A, B and C are traditional classful categories; modern networks mostly use CIDR.

2. Calculate Network Address, Broadcast Address and Valid Host Range

Given:

- IP address: 192.168.10.45
- Subnet mask: 255.255.255.0
- CIDR notation: /24

Step 1: Convert the subnet mask into binary

255.255.255.0

11111111.11111111.11111111.00000000

There are 24 network bits and 8 host bits.

Step 2: Calculate the network address

Perform a bitwise AND between the IP address and subnet mask.

IP: 192.168.10.45

Mask: 255.255.255.0

Network: 192.168.10.0

The last octet becomes 0 because all 8 host bits are zero.

Network address: 192.168.10.0

Step 3: Calculate the broadcast address

Set all host bits to 1. The last octet becomes 255.

Broadcast address: 192.168.10.255

Step 4: Calculate the valid host range

Number of host bits = 8

Total addresses:

2^8=256

Usable host addresses:

256-2=254

We subtract 2 for the network and broadcast addresses.

Final answers:

Item	**Answer**
Network address	192.168.10.0
Broadcast address	192.168.10.255
First valid host	192.168.10.1
Last valid host	192.168.10.254
Total usable hosts	254

3. Subnetting Plan for a Food Delivery Startup

Given network: 10.0.0.0/24

We need 4 separate subnets for Delivery Partners, Restaurants, Customers and Admins.

Step 1: Calculate the required subnet bits

2^n >=4

2^2=4

We borrow 2 bits from the host portion.

New prefix:

24+2=26

New subnet mask: 255.255.255.192

Block size:

256-192=64

Therefore, the subnet network addresses increase by 64.

Step 2: Assign the subnets

Group	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1–10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65–10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129–10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193–10.0.0.254	10.0.0.255

Each /26 subnet has 64 total addresses and 62 usable host addresses.

This plan divides the network into four equal subnets. In a real startup, customer devices would generally be isolated from internal admin systems, and the subnet sizes could be adjusted based on actual demand.

4. Subnetting a College Network Using 172.16.0.0/16

Given:

- Original network: 172.16.0.0/16
- Required subnets: At least 20

Step 1: Find the number of borrowed bits

We need to find \(n\) such that:

2^n >=20

2^4=16 (not enough)

2^5=32 (enough)

Therefore, we borrow 5 bits.

Step 2: Calculate the new prefix

16+5=21

The new subnet mask is:

255.255.248.0

Step 3: Calculate the number of hosts per subnet

Original host bits = 16

After borrowing 5 bits:

16-5=11{ host bits}

Total addresses per subnet:

$2^{11}=2048$

Usable host addresses:

$2^{11}-2=2046$

Final answer

Item	Result
Original network	172.16.0.0/16
Bits borrowed	5
Number of subnets	32
New CIDR prefix	/21
New subnet mask	255.255.248.0
Total addresses per subnet	2048
Usable hosts per subnet	2046

The block size in the third octet is:

256-248=8

The first few subnet network addresses are:

- 172.16.0.0/21
- 172.16.8.0/21
- 172.16.16.0/21
- 172.16.24.0/21

This provides 32 subnets, which is enough for the college's requirement of at least 20 buildings.

5. Why Is 192.168.1.300 Invalid?

The IP address 192.168.1.300 is invalid because an IPv4 address contains four octets, and each octet must have a value between 0 and 255.

The last octet, 300, exceeds the maximum permitted value.

Valid examples in the 192.168.1.0/24 subnet include:

- 192.168.1.1
- 192.168.1.25
- 192.168.1.100
- 192.168.1.254

The usual usable host range is 192.168.1.1 to 192.168.1.254, assuming a /24 subnet. The network address is 192.168.1.0, and the broadcast address is 192.168.1.255.

Conclusion: The device must be assigned a valid IP address within its subnet, with no octet exceeding 255.